

HÉLDER GONÇALVES

Lead Software Engineer

helder@hgoncalves.uk
+351 931 172 969

linkedin.com/in/helderjgoncalves
github.com/helderjgoncalves
hgoncalves.uk

PROFILE

Over 10 years building developer tools used daily by thousands of enterprise engineers. Nearly all of it on one product — the OutSystems IDE — across four eras: its migration off Windows-only WPF onto web technologies, the reliability turnaround that followed, its adaptation into ODC Studio for the company's cloud platform, and now Mentor, the platform's agentic AI, inside that visual IDE.

What stayed constant is changing a live product without interrupting the people who depend on it: replacing it underneath them a piece at a time behind feature toggles, and building the telemetry that says what a failure actually costs them.

WORK EXPERIENCE

Lead Software Engineer

Lead SWE (2023 – Present), Senior SWE (2020 – 2023), SWE (2018 – 2020), Junior SWE (2016 – 2018)

outsystems

AI & Technical Direction

- **Mentor in ODC Studio:** Leading the integration of Mentor — the platform's agentic AI — into ODC Studio: a conversational workflow inside the visual IDE, generally available since March 2026.
- **Architecture Guidance:** Guide architecture for two teams — twelve engineers — and advise engineering groups beyond them on design.

Reliability & Quality Engineering

- **Stability & Metrics:** Led a targeted initiative that cut crash ratio by roughly 80%, and built the telemetry that pinpoints high-impact failures — since extended to hangs and freezes.
- **SRE Practices:** Introduced Site Reliability Engineering practices and dark launches, so new features reach production switched off and roll out gradually.

Developer Experience

- **Design System Modernization:** Led the implementation of the 2026 modernization that took a company-wide Design System live — setting up its infrastructure, and sending designs back to UX for rework when they would not hold up in code.
- **Granular Merge:** Made merge much more granular, so edits to different parts of the same screen no longer conflict. A team initiative added text merge for what remains — from fully manual to hands-off most of the time, with a redesigned UI when a human is still needed.
- **Desktop Notifications:** Brought desktop notifications into the IDE from scratch, so it can reach developers when it is not the focused window.

Technical Architecture & Cloud Integration

- **Cloud Platform Client:** Re-architected the desktop IDE's client integration with the OutSystems cloud platform for ODC Studio, moving it off a single backend dependency onto distributed services.
- **Incremental Platform Migration:** The WPF-to-web migration shipped component by component behind feature toggles — a React UI in an embedded WebView — so the product kept shipping instead of forking into a rewrite. Multi-year team effort; that unlocked macOS support.

React TypeScript .NET AWS Snowflake

Software Developer Internship

January – August 2016

TECHNISCHE UNIVERSITÄT DARMSTADT

Selected for an Erasmus+ exchange at TU Darmstadt's Institute for Scientific Computing, developing optimised software modules for post-quantum cryptography research, including an Arithmetic Multiple Precision library that outperformed the GMP library in key operations.

Parallelised and optimised lattice basis reduction algorithms (LLL and Qiao's Jacobi method) to accelerate cryptographic computations.

Work formed the basis of my MSc thesis at Universidade do Minho.

C | Intel Intrinsics | OpenMP | Assembly

Summer Internship in Advanced Computing Research

July – August 2015

TEXAS The University of Texas at Austin

Built a multithreaded C++ framework for the **GUIDE** project (Graph Unified Irregular Distributed Environment), enabling parallel and distributed operations over complex graph structures.

C++ | Multithreading

EDUCATION

BSc. and MSc. in Informatics Engineering

2011 – 2016

Universidade do Minho

Relevant Coursework
Parallel and Distributed Computing, Distributed Systems & Cryptography, Computer Graphics.

Thesis
Towards an efficient lattice basis reduction implementation — advised by Prof. Alberto Proença and Artur Mariano.

SIDE PROJECTS

Self-Hosted Home Lab — Personal infrastructure (ongoing)

NAS-based private cloud serving the whole family — photos, files and media (Immich, OpenCloud, Plex), running in Docker. No inbound ports: traffic arrives through Cloudflare Tunnel and a Traefik reverse proxy, with zero-trust access at the edge (OAuth, mTLS, service tokens), DNS over HTTPS, and layered monitoring with alerting. The whole stack is declared in Git, so it redeploys onto fresh hardware.

Docker | Cloudflare Zero Trust | Traefik | mTLS | OAuth | DNS over HTTPS | Configuration-as-Code

LANGUAGES

Portuguese (native), English (fluent)

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C01) | Amazon Web Services
- Certified SCRUM Developer (CSD) | Scopphu

LEADERSHIP & AWARDS

- Top Performer Award | OutSystems — Company-wide recognition for delivering high-impact technical results